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Subject: Blockchain Fundamentals Date: 25-04-26

Assisgnment-1

Question: Write about Blockchain and its use in details

Ans:

Blockchain is a type of distributed digital ledger that records transactions across many computers in a way that makes the data secure, transparent, and nearly impossible to alter. Instead of relying on a central authority (like a bank or government), blockchain uses a ***decentralized network*** where every participant (node) has access to the same data.

Each record in a blockchain is called a ***block***, and these blocks are linked together in chronological order, forming a ***chain***. Once a block is added, it cannot be changed without altering all subsequent blocks, which requires consensus from the network..

Key Features of Blockchain

1. ***Decentralization-***
   No single entity controls the system. Data is distributed across multiple nodes.

2. ***Transparency***
   Transactions are visible to all participants in the network (depending on the type of blockchain).

3. ***Immutability-***
   Once recorded, data cannot be easily changed or deleted.

4. ***Security-***
   Uses cryptographic techniques to secure transactions and identities.

5. ***Consensus Mechanisms-***
   Methods like Proof of Work or Proof of Stake ensure agreement among participants

How Blockchain Works

1. A transaction is requested.

2. The transaction is broadcast to a network of computers (nodes).

3. Nodes validate the transaction using algorithms.

4. Verified transactions are grouped into a block.

5. The block is added to the existing blockchain.

6. The transaction is completed and permanently recorded.

Types of Blockchain-:

Public Blockchain: Open to anyone (e.g., Bitcoin, Ethereum)

Private Blockchain: Controlled by a single organization

Consortium Blockchain: Controlled by a group of organizations

Hybrid Blockchain: Combination of public and private

Uses of Blockchain-:

1. Cryptocurrencies

The most well-known use is digital currencies like Bitcoin and Ethereum, which allow peer-topeer financial transactions without banks.

2. Smart Contracts

Programs that automatically execute agreements when conditions are met. For example, on Ethereum, contracts can run without intermediaries.

3. Supply Chain Management

~~Blockchain tracks go~~ods from origin to delivery, improving transparency and reducing fraud.

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[4]Stores patient records securely, ensuring privacy while allowing controlled access to doctors and. Healthcare hospitals.

[5]Enables secure and transparent voting, reducing election fraud. . Voting Systems

[6]Speeds up cross-border payments and reduces transaction costs by removing intermediaries.· Banking and Finance

[7]Users can control their identity data without relying on centralized authorities.· Digital Identity

[8]Blockchain enables ownership of digital assets like art, music, and collectibles· NFTs (Non-Fungible Tokens)

Assisgnment-2

~~Question:~~***Explain any two topics from the syllabus in your words andexplainhow you are going to use the technology in real world.***

Ans:

1***Explanation (in simple words):***.

Signature Scheme (Digital Signature)

A ***signature scheme*** is a method used in blockchain to prove that a message or transaction is genuine and has not been altered. It works using two keys:

Private key → used to sign the transaction
Public key → used by others to verify it

When someone sends data, they sign it with their private key. Anyone can check the authenticity using the public key. This ensures:

The sender is real (authentication)
The data is unchanged (integrity)

For example, in Bitcoin, every transaction is digitally signed before being accepted into the network.

How I will use it in real life:

I would use signature schemes in:

Online document verification (certificates, agreements)
Secure payments to prevent fraud
Login systems without passwords (using cryptographic identity)

For instance, if I run a business, I can use digital signatures to sign contracts with clients so that no one can deny or alter the agreement later

2***Explanation (in simple words):***. Tokenized Blockchain

A ***tokenized blockchain*** is a system where ***digital tokens*** are used to represent value or ownership. These tokens can stand for:

Money (cryptocurrencies like Ethereum)
Assets (land, gold, shares)
Digital items (art, music, NFTs)

These tokens can be transferred easily between users on the blockchain, without needing banks or middlemen.

For example, Ethereum allows creating tokens that represent different types of assets and rights.

How I will use it in real life:

Iwould use tokenized blockchain in:

Investment systems: allowing people to invest small amounts in big assets like property
Loyalty programs: giving customers tokens instead of reward points
Digital ownership: selling or buying digital products (art, courses, music)

For example, if I start a business, I can issue tokens to customers as rewards, which they can later use for discounts or trade with others

Case study

Case Study: Blockchain in Food Supply Chain (Secure & Tokenized System)

Introduction

The food industry often faces problems like ***lack of transparency, data tampering, and insecure payments***. To address these issues, companies are adopting blockchain platforms such as Ethereum to build secure and transparent systems.

Solution Overview

1. Tokenized Blockchain

In this system, products and rewards are converted into ***digital tokens***:

Each product batch is tokenized for tracking
Customers receive tokens as loyalty rewards
Tokens can be redeemed or reused

Using smart contracts on Ethereum, these processes are automated and transparent.

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2. Signature Schemes

To ensure security, all data and transactions are protected using ***digital signatures***:

Farmers, suppliers, and retailers sign data with private keys
Customers sign payments digitally

This guarantees authenticity and prevents tampering, similar to systems used in Bitcoin.

Working Process

1. Product data is uploaded and digitally signed
2. Each supply chain step is recorded on blockchain
3. Customers scan QR codes to verify product history
4. Payments are securely signed
5. Reward tokens are issued automatically

Benefits

Transparent product tracking
Secure and tamper-proof data
Increased customer trust
Efficient reward system through tokens

Real-World Application

If applied in my own business, I would:

Use blockchain to track product quality
Reward customers with tokens
Secure all transactions using digital signatures